

Luxury Housing Sales Analysis – Bengaluru Dashboard

1. Project Overview

The Bengaluru luxury housing market shows dynamic trends in buyer demand, builder performance, and amenity-driven sales. This project integrates PostgreSQL, Python, and Power BI to analyze approximately 101 K records of residential property transactions, providing a 360° analytical dashboard for actionable insights.

2. Data Pipeline

Step 1 – Data Cleaning & Feature Engineering (Python)

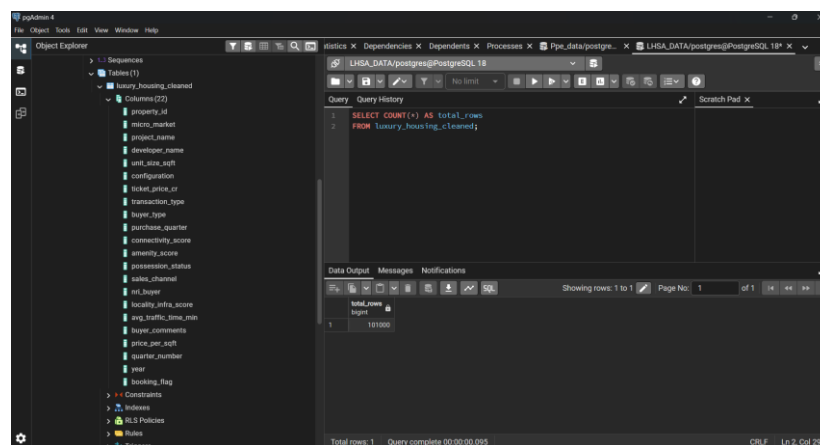
Imported 101 000 raw records into Jupyter Notebook. Cleaned and standardized currency, numeric inconsistencies, and null values. Engineered new features including price_per_sqft, quarter_number, year, and booking_flag.

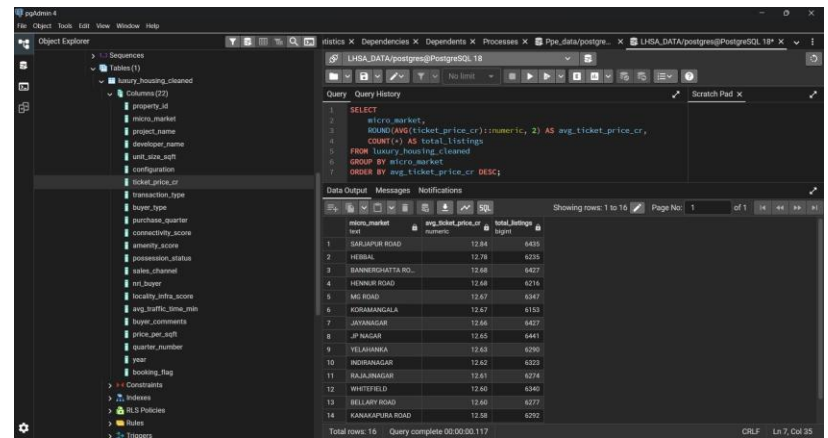
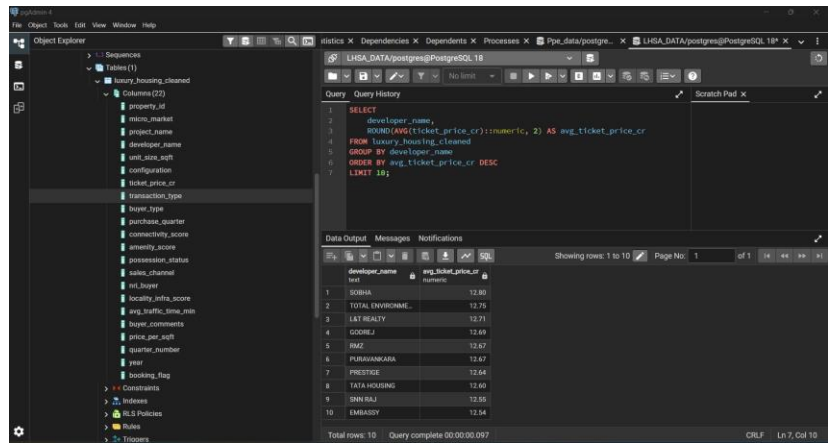
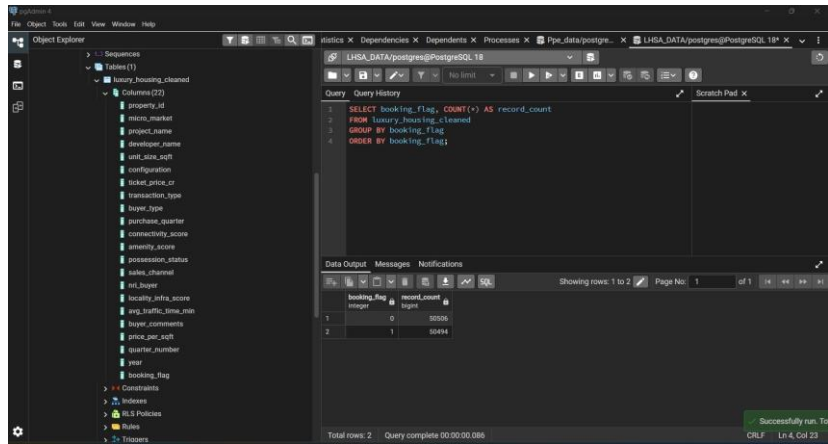
Step 2 – Load Clean Data into RDBMS (PostgreSQL)

After cleaning and transforming the dataset in Python, a connection was established between the Python environment and the PostgreSQL server using the SQLAlchemy library.

The cleaned dataset was then loaded directly into the SQL server, where a new database schema was created to store the refined housing data. This setup ensured a seamless pipeline between Python-based ETL and SQL-based storage for efficient querying and validation

Validated data using pgAdmin queries for row count, booking distribution, and top builders.

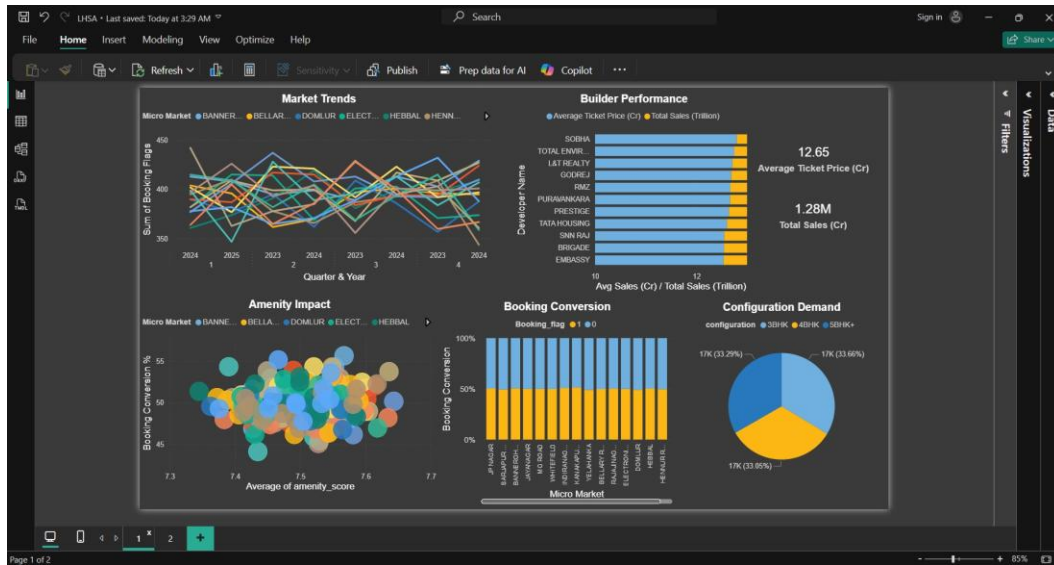




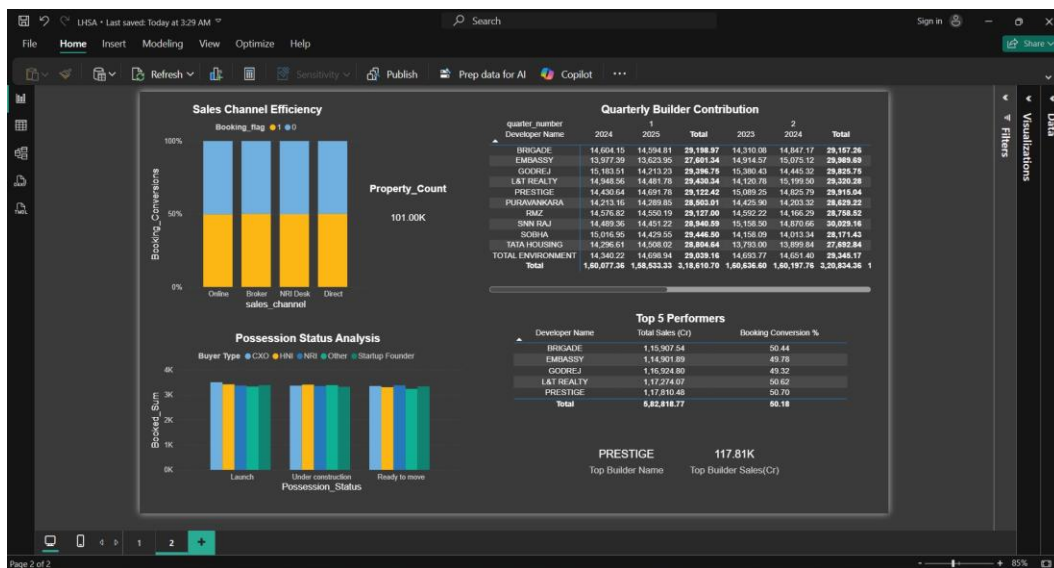
SQL Query Validation Screenshots

Step 3 – Visualization & Dashboard (Power BI)

Connected Power BI to PostgreSQL and built an interactive dashboard. Below are the dashboard pages showing builder performance, amenity analysis, booking conversion, and configuration insights.



Power BI Dashboard Page 1 View



Power BI Dashboard Page 2 View

3. Technologies Used

Data Cleaning & ETL	Python (pandas, sqlalchemy)
Database	PostgreSQL (pgAdmin 4)
Visualization	Power BI

4. Conclusion

The Luxury Housing Sales Analysis – Bengaluru Dashboard enables data-driven understanding of the city's premium real-estate market, highlighting the impact of amenities, builder performance, and booking behavior.